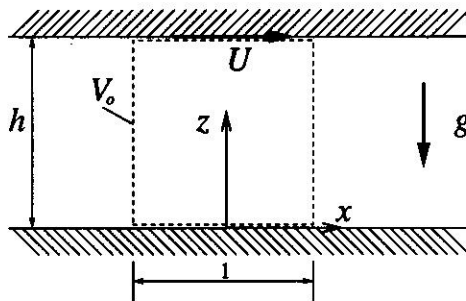


NS4

Consider the steady planar motion of a perfect liquid of density ρ , viscosity μ , thermal conductivity k , and specific heat c in the gap of thickness h that forms between two infinite horizontal walls. The upper wall, at temperature T_u , moves parallel to itself with velocity U , while the lower wall remains at rest, at temperature T_l .

1. Determine the velocity components $v_x(z)$ and $v_z(z)$, as well as the accompanying pressure field $p(x, z)$, assuming that the pressure on the upper wall has the constant value $p = p_u$.
2. Calculate the volume flux for the flow established in the gap.
3. Obtain the pressure distribution $T(x, z)$.
4. Compute the force per unit surface acting on the upper wall as well as the power needed to move it.
5. Determine the conductive heat flux from the fluid to the upper and lower walls.
6. Use the fixed control volume V_0 indicated in the figure to show that the power employed to move the upper wall is dissipated inside the liquid, generating an excess internal energy that is transferred by heat conduction to the walls.



1) $\frac{\partial v_x}{\partial x} + \frac{\partial v_z}{\partial z} = 0$ $\boxed{v_z = \text{const} = 0}$ $\leftarrow z=0, h: v_z=0$

z-Mom: $0 = -\frac{\partial p}{\partial z} - \rho g \Rightarrow \boxed{p + \rho g z = f(x)} \leftarrow z=h, p=p_u$

x-Mom: $0 = -\frac{\partial p}{\partial x} + \mu \frac{\partial^2 v_x}{\partial z^2} \Rightarrow \boxed{v_x = C_1 + C_2 z = \frac{U}{h} z}$ $\leftarrow z=0, v_x=0; z=h, v_x=U$

2) $\vec{Q} = \int \vec{v} \cdot \vec{n} d\sigma = \int_0^h \frac{U}{h} z dz = \boxed{\frac{U h}{2}}$ 3) $0 = \phi_r + k \frac{\partial^2 T}{\partial z^2}$, $\phi_r = \mu \left(\frac{\partial v_x}{\partial z} \right)^2 = \mu \frac{U^2}{h^2}$

4) $\vec{n} = -\vec{e}_z$ $\boxed{-p \vec{n} + \vec{\tau} \cdot \vec{n} = p_u \vec{e}_z + \begin{bmatrix} 0 & 0 & \mu \frac{\partial v_x}{\partial z} \\ 0 & 0 & 0 \\ \mu \frac{\partial v_x}{\partial z} & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} = p_u \vec{e}_z - \mu \frac{U}{h} \vec{e}_x}$ $T = B_1 z + B_2 = \frac{\mu}{k} \frac{U^2}{h^2} \frac{z^2}{2} \Rightarrow \boxed{T = T_l + (T_u - T_l) \frac{z}{h} + \frac{\mu U^2}{2k} \left(\frac{z}{h} - \frac{z^2}{h^2} \right)}$ $z=0: T=T_l; z=h: T=T_u$

5) $\dot{W} = \dot{W}_{w \rightarrow l} = -\dot{W}_{l \rightarrow w} = \mu \frac{U^2}{h}$, $\dot{W}_{l \rightarrow w} = - \int_{\Sigma_w} p \vec{n} \cdot d\vec{\sigma} + \int_{\Sigma_w} \vec{v} \cdot \vec{\tau} \cdot d\vec{\sigma} = -\mu \frac{U^2}{h}$

6) $\frac{d}{dt} \int_{V_0} \rho \frac{v_x^2}{2} dV + \int_{\Sigma} \rho \frac{v_x^2}{2} \vec{v} \cdot d\vec{\sigma} = - \int_{\Sigma} p \vec{n} \cdot d\vec{\sigma} + \int_{\Sigma} \vec{v} \cdot \vec{\tau} \cdot d\vec{\sigma} + \int_{V_0} \rho \vec{g} \cdot dV - \left[- \int_{V_0} p \vec{\nabla} \cdot dV + \int_{V_0} \phi_r dV \right]$ $\int \phi_r dV = \mu \frac{U^2}{h^2} h$

$\frac{d}{dt} \int_{V_0} \rho \frac{v_x^2}{2} dV + \int_{\Sigma} \rho \frac{v_x^2}{2} \vec{v} \cdot d\vec{\sigma} = - \int_{\Sigma} \vec{q} \cdot \vec{n} d\sigma + \left[\int_{V_0} \phi_r dV - \int_{V_0} p \vec{\nabla} \cdot dV \right]$

$-(Q_u + Q_l) + \int \phi_r dV = 0 \Rightarrow \boxed{\int \phi_r dV = Q_u + Q_l}$

5) $\vec{n} = \vec{e}_z$ $\dot{Q}_u = \vec{q} \cdot \vec{n} = -k \nabla T \cdot \vec{e}_z = -k \frac{\partial T}{\partial z} = -k \frac{(T_u - T_l)}{h} + \mu \frac{U^2}{2h}$

$\vec{n} = -\vec{e}_z$ $\dot{Q}_l = \vec{q} \cdot \vec{n} = k \nabla T \cdot \vec{e}_z = k \frac{(T_u - T_l)}{h} + \mu \frac{U^2}{2h}$